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Hierarchical Task Analysis

a. Identify the functions (primary tasks) to be performed in using this product. There should be at most three or four, depending on the product; but each tool you have should have at least two functions. For each function, determine the user/product performance requirements in using the hand tool. See Table 1 for an example.

Tasks to be performed:	Use/Product Performance Requirements
Insert our strainer device onto the pot	<1 minute to complete
Insert pot handle cover onto the handle	<2% error rate
Pour the water through the strainer into the sink	>95% can properly strain the water without spilling any pasta.
Remove the strainer and handle cover from the device	<1 minute to complete

b. For each of your functions, conduct a Hierarchical Task Analysis (HTA), as briefly discussed in class and as described in your textbook. In essence, thoroughly outline the use of your product. Determine what information is needed for each task/subtask associated with carrying out that subtask. Do a library or other search to find an example HTA.

Function	Task and Subtask	Identify Critical Elements			
		Cognitive	Perceptual	Motor	Stress
Insert our strainer device onto the pot	1.Identify placement: -Recognize pot type and handle layout -Determine correct side	Orientation judgment	Visual alignment	Handing devices	Time pressure, steam exposure

	2. Stretch and secure strainer: -Stretch silicone ring/edge -Lock onto rim	Check symmetry	Fit confirmation	Grip, stretching force	Risk of slipping or touching hot surface
Insert pot handle cover onto the handle	1. Check handle: -Assess temperature	Decision-making	Temperature awareness	\	\
	2. Fasten cover: -Slide/Wrap securely	Confirm secure placement	Alignment and fit	Fine hand control	\
Pour the water through the strainer into the sink	1. Lift pot: -Grip handles -Move to the sink	Stability control	Pot positioning	Lifting, balance	Risk of spill, weight load
	2. Drain water: -Tilt pot -Monitor flow	Angle control, timing	Visual tracking of water flow	Smooth wrist/arm motion	Hot stream, fear of spill
Remove the strainer and handle cover from the device	1. Cool device: -Wait until safe	Judgment of safety	Heat detection	\	Anticipation of residual heat
	2. Detach parts: -Release clips -Remove cover	Task sequence awareness	Identify latch/seam	Grip, pull, release motion	Residual heat, slipperiness

c. For each subtask, identify the critical human factors/ergonomic element(s): perceptual, cognitive, (mental), motor (ergonomic), or stress. For each of these critical elements and where possible, identify potential human factors/ergonomics problems (see example in Figure 1). To help you with this, think about the hand tool/system elements of equipment, procedures, humans, and environment. Sample questions are provided in Figure 3, shown on the next page and provided in class.

Function	Observations from questions				Potential HF Problems	Recommendations
	Equipment	Procedures	Humans	Environment		
Insert our strainer device onto the pot	Device must align with pot rim, requires precision.	User must squeeze clips and hold strainer steady. Flip up steam guard.	Requires hand strength, coordination, and correct orientation.	Steam or heat from pot.	Risk of burns from pot and awkward wrist angles.	Add heat-resistant handle grips.
Insert pot handle cover onto the handle	Handle cover must fit snug, and material should be ridged for gripping. .	Completed before pouring water, may require two hands.	Fine motor control ,cognitive step to remember.	Steam and heat from pot.	Forgetting to use the cover, being too slippery and drop the pot.	Ensure universal fit, include some sort of reminder to apply.
Pour the water through the strainer into the sink	Strainer must stay secure, handle must remain cool.	Coordinate pouring at an angle and aiming toward the sink.	Requires upper body strength, grip, and balance. As well as	Steam and wet surfaces.	Wrist/arm strain, slips or spills, burns from steam	Make sure to use steam guard, ergonomic handle angle, non-slip grip, recommend use two hands.

			mental focus.			
Remove the strainer and handle cover from the device	Strainer may be wet or slippery with some heat.	Unclip from pot, place in sink to wash.	Unclip carefully, ensure don't touch pot.	Steam, hot surface from stove.	Burns, slips, cognitive overload from multitasking.	Clear instruction label on how to remove safely.

Hierarchical Task Outline

0. Use the Strainer System to complete the pasta draining task
- 1. Prepare the cookware
  - 1.1 Identify the Pot Type (single handle or dual handle)
  - 1.2 Inspect Rim for residue or obstruction
  - 1.3 Gather strainer device or handle cover
- 2. Attach the Strainer to the pot
  - 2.1 Determine correct orientation relative to handles
  - 2.2 Stretch silicone rim evenly around pot edge
  - 2.3 Secure clips and confirm symmetrical fit
  - 2.4 Visually check seal to prevent leaks or slippage
- 3. Apply handle cover
  - 3.1 Evaluate handle temperature
  - 3.2 Slide or wrap silicone cover fully onto handle
  - 3.3 Verify fim grip cover should not rotate or slip
- 4. Pour and drain water
  - 4.1 Grip handles firmly with both hands
  - 4.2 lift pot from stove and carry to sink
  - 4.3 Tilt pot slowly toward sink and control water flow
  - 4.4 Monitor flow to ensure pasta remains in pot
  - 4.5 Return pot upright when water is fully drained
- 5. Remove and Clean Components
  - 5.1 Allow pot and strainer to cool
  - 5.2 Release the locking clips carefully
  - 5.3 Remove handle cover and avoid any residual heat
  - 5.4 Rinse and dry the components to reuse in the future

d. For each task, answer the questions posed in Figure 2. Each cell (Task to column) will answer one column of questions.

Task	Subtask	Identify Critical Elements	Observations from Questions	Potential HF Problems	Recommendations
Identify placement	Recognize pot type and handle layout	Cognitive/Perceptual	Equipment: Are pot handles clearly visible and distinct in shape? Procedures: Is there clear instruction for which pot to use? Humans: Does the user need prior knowledge to identify handle design? Environment: Is lighting adequate to distinguish handle type?	Poor visibility due to steam or dim lighting may cause confusion between handle types.	Improve lighting near stove; use color-coded or textured handles for quick recognition.
Identify placement	Determine correct side	Cognitive/Perceptual	Equipment: Are both sides identical or labeled for use? Procedures: Is there instruction on which side to use for pouring? Humans: Does user have to recall past experience? Environment: Are labels readable when pot is hot or steamy?	Unclear side identification may cause incorrect placement or spillage.	Add clear directional markings or raised arrows on pot for correct side placement.
Stretch and secure strainer	Stretch silicone ring/edge	Motor/Perceptual/Stress	Equipment: Is silicone flexible and sized correctly for pot? Procedures: Are instructions on	Strain from stretching too tightly; risk of	Provide material with better grip and flexibility; ensure user

			how much to stretch provided? Humans: Is user strength sufficient? Environment: Are hands slippery or wet?	slipping or tearing silicone.	can stretch easily with wet hands.
Stretch and secure strainer	Lock onto rim	Motor/Perceptual	Equipment: Does rim shape support locking? Procedures: Are locking instructions clear? Humans: Is adequate feedback felt/heard when locked? Environment: Is area stable and not obstructed?	Unclear locking feedback may cause detachment during use.	Add tactile/audible 'click' feedback; design universal rim fit.
Check handle	Assess temperature	Perceptual/Mental/Stress	Equipment: Does handle have heat indicator or insulation? Procedures: Is there instruction to check handle before lifting? Humans: Can user judge temperature safely? Environment: Is lighting adequate to see color indicator?	Users may burn hands due to lack of indicator or gloves.	Add built-in heat indicator or recommend heat-resistant gloves.
Fasten cover	Slide/Wrap securely	Motor/Perceptual	Equipment: Is cover flexible and easy to slide? Procedures: Are clear steps given for fastening? Humans: Is motion intuitive? Environment: Are	Cover may not seal properly, causing leaks or instability.	Simplify locking design; use grip tabs for secure fastening.

			surfaces clean and dry?		
Lift pot	Grip handles	Motor/Stress	Equipment: Are handles ergonomically shaped and insulated? Procedures: Is there warning to test weight? Humans: Is user’s grip strength sufficient? Environment: Is space around stove clear?	Slippery or hot handles increase burn and drop risk.	Add anti-slip grips; ensure handle spacing fits all hand sizes.
Lift pot	Move to the sink	Motor/Stress	Equipment: Is pot balanced when lifted? Procedures: Is path to sink clear? Humans: Is user posture stable? Environment: Are floors dry, no obstructions?	Risk of spills or dropping pot during transfer.	Encourage slow, stable movement; clear sink area beforehand.
Drain water	Tilt pot	Motor/Perceptual	Equipment: Is spout design adequate to control flow? Procedures: Are instructions given on tilt angle? Humans: Can user control heavy load safely? Environment: Is sink height comfortable?	Sudden outflow can cause splashing or burns.	Design strainer lip for steady flow; ergonomic handle positioning.
Drain water	Monitor flow	Cognitive/Perceptual	Equipment: Is the strainer visible enough? Procedures: Does user know when draining is complete? Humans: Can user maintain visual	Poor visibility may cause incomplete draining or spillage.	Use transparent or marked strainer; improve visibility of draining area.

			focus amid steam? Environment: Is lighting strong enough?		
Cool device	Wait until safe	Cognitive/Stress	Equipment: Is there a visual indicator for safe temperature? Procedures: Is cooling time specified? Humans: Does user know how long to wait? Environment: Is ambient temperature affecting cooling?	User may handle too soon, causing burns.	Add color-change heat indicator; specify safe cooling time in manual.
Cool device	Detach parts	Motor/Perceptual	Equipment: Are release clips easy to operate? Procedures: Are instructions for release steps clear? Humans: Can user perform task without excessive force? Environment: Is workspace stable and dry?	Clips may jam or require too much force when hot.	Redesign release mechanism for single-hand use; ensure safe operation after cooling.
Cool device	Remove cover	Motor/Perceptual	Equipment: Is cover easy to grasp and remove? Procedures: Are steps for removal clear? Humans: Is motion intuitive and low effort? Environment: Is steam still present?	Burns or slippage during removal.	Add steam vent and grip tab; recommend waiting for full cooldown.

Discussion/Issues to Consider: As you complete the first stage of your product human factors engineering/ergonomics-based design or redesign, discuss some of the issues/things your team will take into consideration in the design or redesign. Have you thought of these things prior to conducting the task analysis?

As we move into the first stage of our product’s human factors based redesign, our team will focus on several key considerations. First, we need to ensure that the design accommodates a wide range of users, accounting for differences in hand size, grip strength, and experience level. Ergonomic comfort and ease of use are top priorities, especially when the product involves repetitive or heat-related tasks. We’ll also pay attention to visibility and feedback, making sure users can clearly identify components, feel when parts are securely locked, and receive clear cues that it’s safe to handle.

Environmental factors like lighting, surface wetness, and temperature will influence both safety and usability, so the design must perform well under real kitchen conditions. Durability and maintenance are also important materials should be easy to clean, resistant to heat, and long-lasting.